From Visualization to Source Analysis

This group discussion helped me understand that digital history tools play a role throughout source collection, organization, analysis, and communication. Both classmates' presentations made historical questions easier for me to understand and encouraged me to pay attention to the research process behind the visuals. This led me to think further: how can we use digital tools to ask questions, analyze sources, and involve the public in sharing historical knowledge?

The animated map that Wang Yihan presented on her webpage quickly made my previously abstract understanding clearer. I think QGIS is suitable for her project because spatial layers can clearly show regional differences in the placement of *karun* (frontier posts) and changes over time. This can help readers understand the relationship between space and frontiers. If sufficient historical sources and further explanation become available, StoryMap could also bring together maps, original texts, and background information, providing an interactive way to read for people unfamiliar with specialist software. These two tools can support spatial analysis and public understanding respectively.

My main suggestion is that she extend her use of digital tools to source analysis. Her discussion of Russian expansionist intentions and the expanding and contracting areas shown in the animation made me ask: besides territorial expansion, could other factors have influenced changes in the frontier posts? For example, could local residents' responses to policies of governance have affected the addition or removal of posts? This could be a hypothesis to test. If relevant records can be found, she could first use Excel to organize dates, locations, changes in the posts, and responses to policies, and use pivot tables to compare their distribution. When there are enough sources and the categories are comparable, she could then use SPSS for cross-tabulation or correlation analysis to identify connections worth returning to the historical sources to explain.

This analysis could also support further discussion of the meaning of frontier governance. However, residents' attitudes must be identified from specific sources and cannot be assigned arbitrary values. Correlation also cannot directly prove causation. The placeholder records in the current animation are not yet sufficient to support these analyses, so it would be better to begin with a small number of verifiable cases. At the same time, silence in historical sources cannot be coded as residents having no opinions. Preserving the original wording, the perspectives of the sources, and points of uncertainty can make the analysis more reliable. It can also prevent the perspective of state archives from obscuring local society, allowing the digital work to retain the complexity of frontier relations.

Hao Xuanying's presentation made me think that the public value of historical research also includes passing on culture, while heritage protection requires public understanding, recognition, and participation. Digital tools offer new ways to support

this. In the panoramic display discussed in class, the 360-degree view expanded visual access, but she hoped that visitors could also ask questions, choose routes, and give feedback on their experiences. Her project therefore raises a question worth exploring further: how can digital technology turn opportunities to view into opportunities to participate in and understand history?

However, her idea of collecting visitor feedback, which she mentioned during our conversation, also made me question how the tools matched the goals. Scrollytelling is suitable for explaining history step by step, but real-time feedback also requires information collection, storage, and analysis. As I understand it, this idea could develop into a mini-program prototype combining dynamic visitor guidance, route selection, experience feedback, and visitor flow information. I suggest first choosing a few locations to test two functions—visitor guidance and feedback—and then using Excel to summarize choices and suggestions. If the sample allows, SPSS could be used to compare experiences across different routes. This would help test how feedback could improve visits without having to build a complete platform at the beginning. Visitor flow information would require a separate, reliable source; the number of people submitting feedback cannot serve as a substitute.

This suggestion also comes from my previous experience in a project promoting historical memory in Chinatown. Besides collecting historical sources, we conducted field visits, interviewed shop owners, and collected and analyzed opinions through an online questionnaire. We eventually created an interactive mini-program model. The model combined route selection, introductions to attractions, business hours and other business information, and suggestion submissions, using GPS, data analysis, and basic modeling tools. This experience helped me realize that members of the public are both readers of historical narratives and potential contributors of local memories and user experiences. I have shared these experiences with Hao Xuanying, hoping they can offer a useful reference for collecting visitors' impressions.

Compared with organizing institutional documents, the challenge of this additional direction in her project may lie in understanding people's feelings. Interviews can preserve specific experiences, while questionnaires make it easier to compare opinions, so the two methods can complement each other. However, people willing to respond online may not represent all visitors. It is therefore necessary to reach different groups as far as possible and allow open-ended answers. When collecting feedback, she should explain its purpose and obtain consent, and remove personally identifying information before making it public. If location tracking is involved, people should be allowed to decline. Language, devices, and internet access also need to be considered; otherwise, a platform that emphasizes interaction may still exclude some members of the public.

Bringing these reflections back to my Dunhuang project, I became more clearly aware that source organization and analysis deserve as much attention as the final network graph. Gephi can connect people, institutions, and transfer events found in scattered

records, helping me compare the Stein route with the Qing government's route to Beijing. Drawing on Wang Yihan's approach, I can first compare the stages of handling, event types, and distribution of sources along the two routes, and then ask how their structural differences developed. The current sample is very small, so descriptive organization and comparative reading of historical sources are appropriate. Node size also needs to be explained in relation to how detailed the records are and cannot be directly equated with historical power. Hao Xuanying's project inspired me to include public understanding and feedback in the design of my Dunhuang project's output. I could first use a single chain of transfers to explain the differences between "acquisition," "transport," and "institutional receipt," before presenting the full network. I could then ask test readers to find the evidence supporting the connections and identify points that are difficult to understand. This interaction could help improve the historical explanation. Meanwhile, the source table should preserve the explorers' original wording and perspectives, allowing the roles of Wang Yuanlu, translators, and other people who handled the manuscripts to be shown.

This group discussion helped me understand that tool selection should begin with specific research needs. Maps, statistical analysis, interview organization, and interactive platforms can play roles at different stages. Especially at the current stage, when collecting and organizing sources is relatively difficult, it is worth investing effort in the ability of digital technology to help us organize evidence, compare sources, and develop explanations. Connecting these analyses with outputs for the public can more fully demonstrate the research significance and social value of digital history.